Exercise (Sequence Diagram)

1. **ATM System – Withdraw Cash**

- The user inserts their ATM card into the machine.
- The user enters their PIN for authentication.
- The system verifies the PIN and grants access.
- The user selects the withdrawal option and enters the amount.
- The system checks the account balance.
- The system dispenses cash if sufficient funds are available.
- The system updates the account balance and prints a receipt.

2. **Online Shopping System – Place Order**

- The customer browses products on the platform.
- The customer selects items and adds them to the cart.
- The customer proceeds to checkout.
- The system requests shipping and payment details.
- The customer confirms the order and makes payment.
- The system processes the payment and confirms the order.

3. Ride Booking System – Book a Ride

- The user enters pickup and destination locations.
- The system displays available drivers and fare estimates.
- The user selects a ride and confirms the booking.
- The system assigns a driver to the user.
- The system notifies both the driver and the user.
- The ride is tracked until completion.

4. Hospital Management System – Book Appointment

- The patient logs into the system.
- The patient searches for a doctor by specialty.
- The patient selects a suitable time slot.

- The patient submits the appointment request.
- The system checks availability and confirms the booking.
- The patient receives appointment confirmation.

5. Movie Ticket Booking System – Book Ticket

- The user selects a movie and showtime.
- The system displays available seats.
- The user selects preferred seats.
- The user enters payment details.
- The system processes the payment.
- The system generates and sends the ticket.

6. Banking System – Transfer Money

- The user logs into their account.
- The user selects the fund transfer option.
- The user enters recipient details and amount.
- The user confirms the transaction.
- The system verifies and processes the transfer.
- The system updates balances and sends confirmation.

7. Airline Reservation System – Book Flight

- The customer searches for flights.
- The system displays available flight options.
- The customer selects a flight.
- The customer enters passenger details.
- The customer makes payment.
- The system confirms booking and issues an e-ticket.

8. Parking Management System – Reserve Parking

- The driver checks for available parking slots.
- The system displays available options.
- The driver selects a parking slot.

- The driver confirms the reservation.
- The system allocates the slot.
- The system sends booking confirmation.

9. Food Delivery System – Order Food

- The user browses restaurants and menus.
- The user selects food items and adds them to the cart.
- The user confirms the order and makes payment.
- The system sends the order to the restaurant.
- The system assigns a delivery rider.
- The user tracks the order until delivery.

10. Payroll System – Generate Salary

- The HR manager enters employee details.
- The system calculates salary based on rules.
- The system includes bonuses and deductions.
- The HR manager verifies the data.
- The system generates payslips.
- The system processes salary payments.